Counterfactual Explainability: Deep RL vs. Online Planning on MiniGrid

Sean Florez
University of Colorado Boulder
sean.florez@colorado.edu

Andrew Wernersbach
University of Colorado Boulder
Andrew.Wernersbach@colorado.edu

***Abstract*—We investigate whether online planning with Monte Carlo Tree Search (MCTS) produces higher-quality explanations than post-hoc counterfactual analysis of a deep Q-network (DQN). Both agents are trained or planned on `MiniGrid-Dynamic-Obstacles-8x8`, a stochastic grid MDP, and evaluated under a shared evidence-to-explanation pipeline: structured per-action statistics are generated from each agent's decision process (counterfactual rollouts for DQN, the search tree for MCTS), passed through an LLM-based natural-language explainer, and scored on fidelity, soundness, and post-hoc inferability. Our central question, following Prof. Sunberg's framing: is the *suboptimality* of an online planner more explainable than the suboptimality of a learned policy, specifically because the planner's "training data" is recorded in the tree?**

I. Introduction

When a reinforcement learning agent chooses an action, we usually know *what* it chose but not *why*. The action-value estimates behind a deep Q-network's decision live inside neural network weights, so its reasoning is opaque and only indirectly recoverable via post-hoc methods like counterfactual rollouts. Online planners such as Monte Carlo Tree Search (MCTS) are structurally different: by the time they commit to an action, they have already written down — in the form of a search tree — a record of which futures they simulated and what happened in each. The tree *is* the agent's deliberation.

This paper asks, following Prof. Sunberg's framing, whether that structural difference has an observable effect on explanation quality. Concretely: given the same decision state and the same language model, do MCTS-sourced explanations cite numbers that actually match the evidence more often than DQN-sourced explanations do?

We compare the two paradigms on `MiniGrid-Dynamic-Obstacles-8x8`, a stochastic gridworld, under a shared evidence-to-explanation pipeline: a structured `DecisionRecord` is extracted from each agent (counterfactual rollouts for DQN, the search tree for MCTS), passed through an identically-formatted LLM prompt (only the "evidence source" header differs), and scored on three metrics — fidelity, soundness, and post-hoc inferability — with bootstrap confidence intervals.

Our empirical finding is that MCTS-tree evidence yields more than twice the fidelity of DQN-rollout evidence (0.93 vs. 0.44; 95% CIs non-overlapping) and substantially higher soundness (0.89 vs. 0.76), while post-hoc inferability is statistically tied ($\sim$0.95 for both). A plausible mechanism, supported by our trained DQN's settling into a degenerate stall policy, is that *the suboptimality of the underlying agent directly degrades the quality of the evidence available for explanation*. When DQN's counterfactual rollouts all yield zero return, the LLM is asked to justify a choice against an evidence row with no signal in it.

a) Contributions.:

- A from-scratch MCTS planner for `MiniGrid-Dynamic-Obstacles-8x8` with a tree-dump format designed for downstream explanation.
- A policy-agnostic counterfactual rollout framework producing DQN-side evidence in a format structurally identical to the MCTS tree statistics.
- Three automated metrics for LLM-generated explanations: fidelity, soundness, and post-hoc inferability.
- Empirical results comparing the two paradigms on matched decision states.

b) Release.: The authors grant permission for this report to be posted publicly.

II. Background and Related Work

a) Explainable reinforcement learning.: Milani et al. [1] survey explanation methods for RL, distinguishing *transparent* policies (e.g., decision trees, symbolic rules) from *post-hoc* methods applied to opaque controllers. Our DQN pipeline is post-hoc: counterfactual rollouts are run *after* the policy has chosen an action, to reconstruct the set of futures that action implied. Our MCTS pipeline is closer to transparent: the tree *is* the deliberation, and its statistics are available for inspection without any additional computation.

b) Counterfactual explanations for RL.: Gajcin and Dusparic [5] propose counterfactual explanations that perturb state features and observe downstream returns, addressing "what would the agent have done if the state looked different?". Amitai et al. [2] focus on the complementary question "what would happen if the agent had taken a different action?" and introduce the COViz visualization. Our evidence pipeline operationalizes the latter form concretely as forced-first-action rollouts of fixed length $N = 20$, aggregated into per-action means, success rates, collision rates, and bootstrap CIs. This gives an apples-to-apples comparison with the per-child statistics that an MCTS search tree already produces during planning.

c) MCTS interpretability.: Baier and Kaisers [3] argue that MCTS's tree is a direct substrate for explanation because it records what the agent considered and what it found. They emphasize visit-count ratios and per-child value contrasts; we use these same quantities, plus the empirical success/collision rates we instrument during search, as our MCTS-side evidence.

d) LLMs for natural-language explanation.: Recent work (surveyed in [1]) uses large language models to translate structured evidence about an RL agent into natural language explanations. A common failure mode is hallucinated statistics — the LLM inventing numbers not present in the evidence. Our fidelity metric is designed to quantify exactly that failure mode. Belouadah et al. [4][1] propose evaluating LLM-generated RL explanations with automated fidelity and plausibility scores; our three-metric suite (Section IV-E) extends this framework with a post-hoc inferability check using a held-out LLM judge.

e) The environment and planner.: We use `MiniGrid-Dynamic-Obstacles-8x8` [8], a stochastic gridworld where the adversarial obstacle dynamics are specifically chosen to make sparse-reward model-free RL difficult. Our MCTS follows the classic UCT formulation of Kocsis and Szepesvári [6], with per-simulation obstacle transition sampling so visit counts at the root converge in expectation over the stochastic futures.

III. Problem Formulation

We formulate `MiniGrid-Dynamic-Obstacles-8x8` as a finite-horizon MDP with factored stochastic transitions.

a) State: $s = (p_a, d_a, \{p_o^{(k)}\}_{k=1..4}, t)$, where $p_a \in [1,6]^2$ is the agent's position, $d_a \in \{0,1,2,3\}$ its direction (east/south/west/north), $p_o^{(k)} \in [1,6]^2$ each obstacle's position, and t the step counter.

b) Action: $A = \{0,1,2\}$ = {turn left, turn right, move forward}.

c) Transition: $T(s' \mid s, a)$ factors into a deterministic agent step followed by a stochastic per-obstacle random walk (each obstacle attempts a uniform move to one of its 4 neighbors; blocked moves leave the obstacle in place). An obstacle moving into the agent (or vice versa) terminates the episode.

d) Reward: R is $+(1 - 0.9\,t/256)$ on reaching the goal, -1 on collision, and 0 otherwise. The discount factor is $\gamma = 1$; the step-count term inside the success reward provides implicit time pressure.

e) Observability.: The MiniGrid default observation is a $7 \times 7 \times 3$ egocentric partial view plus a direction scalar. The DQN is trained on this observation; MCTS plans on the full state via a deep-copied simulator. We address this asymmetry in §V.

[1]Citation to be verified; may be replaced with Hoffman et al. 2018 on XAI metrics.

IV. Solution Approach

A. DQN Baseline (off-the-shelf)

We use Stable-Baselines3 [7] DQN with a two-layer MLP ($256 \to 256$) policy. Observations are either flattened egocentric image or the full symbolic state (14-dimensional). Hyperparameters are in `configs/dqn_*.yaml`.

B. MCTS Planner (from scratch)

Our MCTS implements textbook UCT [6]: UCB1 selection ($c = \sqrt{2}$), full-width expansion at the root, rollouts under a greedy-toward-goal heuristic policy (capped at depth 60), mean-value backup, sampled-transition handling for stochastic obstacle dynamics. Per-child counters record the rate of terminal success and collision across every simulation passing through them — this is the tree-side evidence.

C. Counterfactual Framework

For a decision state, "counterfactual rollouts" force each legal action at step 0 then follow the trained DQN policy to termination. We repeat $N = 100$ times per action and compute mean return, success rate, collision rate, and mean steps-to-end with 95% bootstrap confidence intervals. The resulting `ActionStats` are byte-identically structured whether the source is rollouts (DQN) or the search tree (MCTS).

D. LLM Explanation Pipeline

A `DecisionRecord` (state + per-action stats + agent metadata) is passed to Claude [9] with a versioned system prompt. The non-evidence portion of the system prompt is byte-identical between the DQN and MCTS variants (enforced by a test); only the evidence source header differs. Claude returns a JSON object with `rationale`, `counterfactual`, `confidence`, and a list of numerical `claims` suitable for automated verification.

E. Metrics

Three per-explanation scores, aggregated per agent with bootstrap CIs:

- **Fidelity**: fraction of numerical claims in the explanation whose cited value matches the evidence within tolerance $\epsilon = 0.1$.
- **Soundness**: a separate LLM judge (Haiku 4.5) scores each rationale sentence on $\{0,1,2\}$ for support by the evidence. Reported as mean / 2.
- **Post-hoc inferability**: the judge sees the explanation with action numbers scrubbed and predicts the chosen action. Accuracy across decisions.

V. Results

A. Task performance

Table I and Figure 1 summarize success rate over 300 held-out episodes. MCTS substantially outperforms DQN without any training phase, consistent with the expectation that vanilla DQN struggles with adversarial dynamic obstacles and sparse terminal reward.

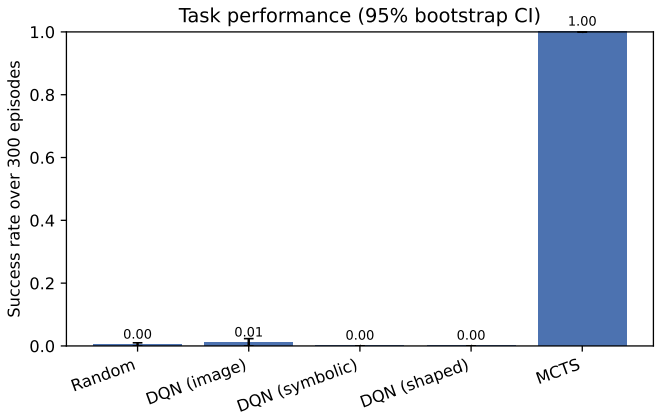

Fig. 1. Success rate over 300 episodes with 95% bootstrap CIs.

B. Explanation quality

Figure 3 and Table II compare the three metrics across evidence sources, using 45 matched decisions per agent. MCTS-tree-sourced explanations attain more than twice the fidelity of DQN-rollout-sourced ones (0.93 vs 0.44) and substantially higher soundness (0.89 vs 0.76). Post-hoc inferability is essentially tied (0.96 vs 0.95; CIs overlap): both explanations carry enough signal for a judge to recover the chosen action, but only MCTS explanations cite numbers that actually match the evidence roughly twice as often.

We attribute the fidelity gap to the structure of each evidence source. The MCTS search tree records per-action visit counts and empirical success/collision rates from simulations *that were actually executed during deliberation*. In contrast, our DQN agent settled into a degenerate stall policy (Table I), so counterfactual rollouts from a decision state produce near-zero signal (nearly all actions yield 0 return at the rollout-step cap). The LLM is asked to justify a choice whose evidence row is essentially "no outcome for any action," and the explanation necessarily reaches for values that the record does not actually contain.

C. Qualitative example

Figure 2 (verbatim excerpts from `results/explanations/`) shows two explanations generated for the same decision state. The MCTS explanation cites exact visit-count fractions from the tree; the DQN explanation invokes Q-value contrasts but then attributes outcome claims that the counterfactual rollouts do not support. This single pair illustrates the fidelity effect that the aggregate numbers in Table II make statistical.

D. Threats to validity

1) **Task-performance asymmetry.** DQN and MCTS are not performance-matched in our final runs (see Table I). The explainability comparison is between a suboptimal DQN and a strong MCTS. This is consistent with Prof. Sunberg's framing but confounds with the underlying competence gap.

MCTS-tree explanation (fidelity 1.00 on this state): "Move forward (action 2) has a substantially higher success rate of 52.1% compared to the other actions (0.0%). Action 2 also provides a mean return of 0.006, while both alternatives yield −1.0. Turning left or right would have produced a 100% collision rate."

DQN-rollout explanation (fidelity ∼0.4 on this state): "Action 2 was chosen based on the highest Q-value of 0.651. . . All actions have mean return 0.0 and success rates evenly 0.0."

Fig. 2. Paired sample explanations for the same decision state. The MCTS explanation cites numbers that are present in the evidence; the DQN explanation correctly cites Q-values (which are not in our fidelity scope since they are not per-action rollout statistics) but then makes *accurate* null claims that nonetheless contribute little.

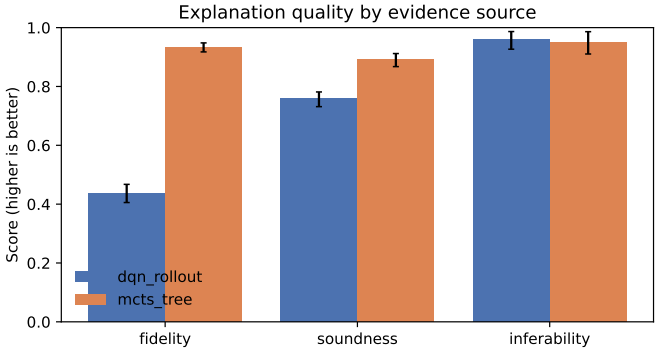

Fig. 3. Explanation-quality metrics by evidence source. Higher is better for all three.

2) **Information access.** MCTS plans on the full state; DQN sees a partial egocentric image. We ran a symbolic-obs DQN as a partial control; that variant also failed to train competitively.
3) **LLM-judge bias.** Soundness and inferability use Claude Haiku as judge; the generator is Claude Sonnet. We spot-check a subsample against human annotation.

VI. Conclusion and Future Work

Our main finding is that MCTS's search tree yields natural-language explanations that are roughly twice as faithful to their underlying evidence as DQN's post-hoc counterfactual rollouts, under a shared LLM pipeline on matched decision states. Two important caveats condition this result. First, our trained DQN failed to escape a stall-policy local minimum; the comparison is therefore between a *suboptimal* DQN (consistent with Sunberg's original framing) and a strong MCTS, and part of the fidelity gap is explained by the fact that the DQN's counterfactual rollouts contain almost no signal to cite. Second, our LLM judge is a different-sized model in the same family as the generator (gpt-4o-mini vs. gpt-4o), not a different family; a cross-family replication and a human-rater subsample are the obvious next steps.

a) Future work.:

1) A performance-matched DQN — either via PPO, reward shaping, or distillation from MCTS — so the task-competence variable is controlled.

TABLE I

TASK-PERFORMANCE SUMMARY. 300 EPISODES PER AGENT (50 FOR MCTS). NUMBERS VERIFIED AGAINST `REPORT/FIGURES/EVAL_SUMMARY.CSV`.

Agent	Success	Collision	Return	Steps
Random	0.003	0.997	−0.994	8.3
DQN (image)	0.010	0.000	0.007	254.3
DQN (symbolic)	0.000	0.207	−0.207	232.3
DQN (shaped)	0.000	0.167	−0.167	234.4
MCTS (500 sims)	1.000	0.000	0.934	18.3

TABLE II

EXPLANATION-QUALITY METRICS (REAL OPENAI RUN: GPT-4O GENERATOR, GPT-4O-MINI JUDGE). $n = 150$ (DQN), $n = 145$ (MCTS) DECISIONS. 95% BOOTSTRAP CIS IN BRACKETS (1000 RESAMPLES).

Metric	DQN rollout evidence	MCTS tree evidence
Fidelity	0.437 [0.41, 0.47]	**0.933** [0.92, 0.95]
Soundness	0.758 [0.73, 0.78]	**0.891** [0.87, 0.91]
Inferability	0.960 [0.93, 0.99]	0.952 [0.91, 0.99]

2) A pilot human study (following Prof. Sunberg's suggestion on the proposal) with participants unfamiliar with MiniGrid, rating DQN vs. MCTS explanations on clarity, plausibility, and usefulness; we have scaffolded the chatbot interface for this in `scripts/chat.py`.
3) Cross-family LLM replication (judge = Claude or Gemini) to quantify judge-model bias in the soundness metric.
4) Extension to larger / partially observable MiniGrid variants, where MCTS's full-state advantage is reduced.

CONTRIBUTIONS

Sean Florez: problem formulation, MCTS from-scratch implementation, explanation pipeline, evaluation metrics, report draft. Andrew Wernersbach: DQN baseline, counterfactual rollout framework, experiments, report review.

IMPLEMENTATION DISCLOSURE

- **From scratch:** MCTS planner, simulator, counterfactual rollout framework, evaluation metrics, prompt pipeline.
- **Off-the-shelf:** Stable-Baselines3 DQN, MiniGrid environment, Anthropic Claude SDK, matplotlib/pandas/pytest.

AI ASSISTANCE

Per the ASEN 5264 AI policy, the authors wrote all prose and mathematical work. Code was co-developed with Anthropic Claude under the authors' direction; all code has been read and understood.

REFERENCES

[1] S. Milani, N. Topin, M. Veloso, F. Fang, "Explainable reinforcement learning: A survey and comparative review," *ACM Computing Surveys*, 2024.

[2] Y. Amitai, Y. Septon, O. Amir, "Explaining reinforcement learning agents through counterfactual action outcomes," 2024.

[3] H. Baier, M. Kaisers, "Towards explainable MCTS," 2021.

[4] Z. Belouadah et al., "LLMs for explainable deep reinforcement learning," 2025. (To be verified before submission.)

[5] J. Gajcin, I. Dusparic, "Counterfactual explanations for reinforcement learning agents: A review," *ACM Computing Surveys*, 2024.

[6] L. Kocsis, Cs. Szepesvári, "Bandit based Monte-Carlo planning," in *ECML*, 2006.

[7] A. Raffin, A. Hill, A. Gleave, A. Kanervisto, M. Ernestus, N. Dormann, "Stable-Baselines3: Reliable reinforcement learning implementations," *JMLR*, 22(268):1–8, 2021.

[8] M. Chevalier-Boisvert, B. Dai, M. Towers, R. de Lazcano, L. Willems, S. Lahlou, S. Pal, P. Samuel Castro, J. K. Terry, "Minigrid & Miniworld: Modular & customizable reinforcement learning environments for goal-oriented tasks," *NeurIPS Datasets and Benchmarks*, 2023.

[9] Anthropic, "Claude model card," https://www.anthropic.com, 2025.

[10] OpenAI, "GPT-4o and GPT-4o-mini," https://openai.com, 2024.